Sprint Progress Report 2

Sprint Overview

During Sprint 2, our team focused on completing the user interface requirements through paper prototyping and organizing development tasks using GitLab issues. We also established a clear work division across frontend, backend, and database components to prepare for implementation.

1. Class Presentation

Our team completed the in-class presentation requirement

2. Work Completed

- Created GitLab issues to break down project tasks into manageable components
- Structured development into three main areas: frontend, backend, and database
- Designed and completed 9 required paper prototype screens based on the UI rubric
- Developed core user flows including login, registration, dashboard, expense tracking, and reporting
- Initiated backend architecture planning, including API and database schema design
- Began frontend scaffolding for login and registration interfaces
- Provisioned a university virtual machine server and initialized a MySQL database with a users table, containing fields for full name, email, password, and user ID, with user ID set as the primary key

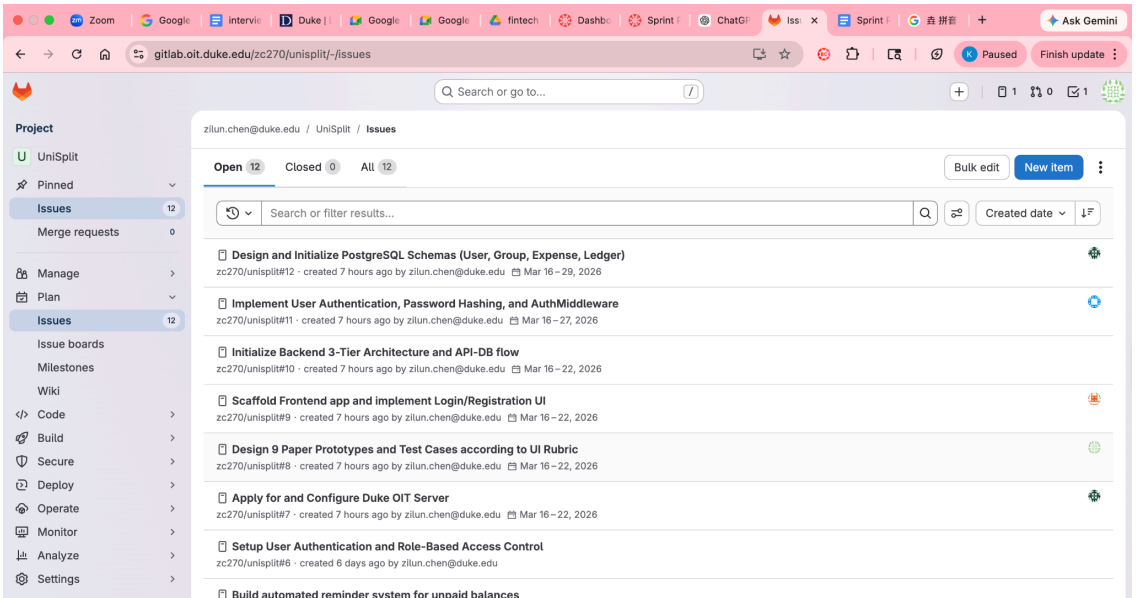
3. GitLab Issues & Work Division

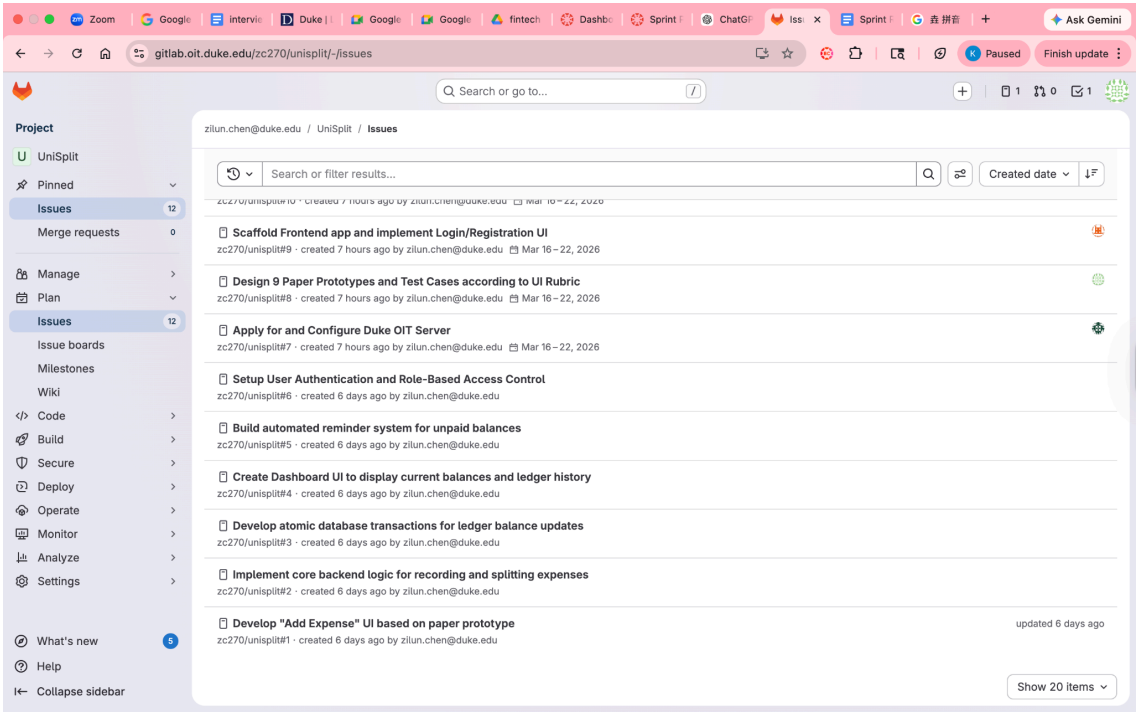
We created and organized GitLab issues to manage tasks and track progress efficiently.

All major features were converted into GitLab issues
Tasks were clearly divided into:

Frontend: UI design, login/registration, dashboard (Chenyao An, Kehan Zhu)
Backend: authentication, API flow, business logic (Jiehe Wu, Zilun Chen)
Database: PostgreSQL schema, transactions, ledger system (Jiakai Lin)

Each team member was assigned one or more issues to ensure accountability





4. Project Artifacts Updates

Our project artifacts were updated to reflect the current progress of the sprint:

- Completed paper prototypes for all required UI screens
- Updated use cases to align with UI functionality
- Improved system design based on feature breakdown (e.g., authentication, expense splitting, reporting)
- Ensured consistency between UI design and planned backend logic

5. Weekly Meeting Summary

Held weekly meeting at Mar 22nd with TA and all team members present
Reviewed paper prototypes and received feedback on UI design
Discussed system architecture and clarified feature scope
Identified priorities for implementation in the next sprint

6. Challenges & Learnings

- Initially, it was challenging to translate abstract requirements into concrete UI screens
- We needed to iterate on prototypes to ensure they matched the rubric and user flow
- Learned how to break down a large system into smaller GitLab issues
- Improved collaboration by clearly separating frontend, backend, and database responsibilities

7. Next Steps (Sprint 3 Plan)

For the next sprint, our team plans to focus on the database, backend foundation and system integration. Specifically, we will:

- refine and finalize the database schema for core entities such as users, groups, expenses, and ledger records
- further break down and define API endpoints for authentication, expense creation, expense
- splitting, balance calculation, and transaction history
- clarify the interaction between frontend pages and backend services
- continue implementation of backend logic and database integration
- begin connecting frontend screens with the corresponding API structure